AI-Driven Platform Guiding Paddy Farmers Across the Full Cultivation Lifecycle

Project Id: 25-26J-242

Project Proposal Report

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Declration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my Supervision.

..... 29/08/2025

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Abstract

Agriculture is the backbone of the Sri Lankan economy with paddy field cultivation being the most widely practical farming activity in the around the Sri Lanka But not having technology for paddy field cultivation , So it is most sadness thing about the Sri Lankan technology world . All farmers and new wanted to be farmer with having not expiring knowledge of the paddy yield are conduct the low property from the paddy yield .Our very obedient object also get more yield and farmers having new life with technology changes and adopt the technology world with using our guidance system. There is some despicably in the sector's the signification, paddy framers are faced numerous challenges that have threaten by the productivity, sustinitibility and livelihood. These issues have largely due to the outdated practices and not having including world technology, limited access to the real time data and lack of the advanced technology support for farmers. Traditional farming methods are reliant on human intuition, irregular weather patterns, and generic agricultural calendars like are no longer sufficient in the face of increasing climate variability, soil degradation, and pest attackers. Many small farmers have lack of access to the modern tools and actionable insights, and easy farming technology used for other countries which leads to inconsistent yields, inefficient resource use, and economic hardship. So having all solution in one system is really reliable for agricultural field future knowledge and gain new updated all over the world matching system.

In this chapters all over I would like explain my individual components of most important and all yield property are based on Pest Attack forecasting component. Pest infestations is often detected too late then full of all harvest effect and get low harvest because of this detective is to doing the too late. Can wipe out entire fields, as most farmers have no reliable knowledge about this pest effected time to predict or prevent outbreaks. While having very advanced varieties offer resistance and having new pests and disease can metalize and over certain solutions can lead the clearly resistance develop pests. Pest really hard to manage if we don't do the right method for solution for type of pest behaviors and also without having grown up knowledge of the pest specifies. This system used historical weather data , crop data ,pest infection data records for upcoming pests attack by notify altering framer in advanced it helps farmers to the reduce pesticide overcome with overdue crop damages and profit loss. It must be learn from the past outbreaks data predictive according to the pest movement in crops ages and behaviorists and alert's farmers before it's too late to the do the treatment for the crops. In this case framers can know about what kind of pest attacks coming for what the reasons like weather situation changes and ages of the paddy crop can easily suffering from attacks and also paddy seed types. Humidity level, soil conditions , pest data from past as multiple real- time data using multimodal input allows to the AI predict more accuracy by learning how to different environmental factors contribute to pest outbreak in specific locations and different stages of the paddy growth. This Depth is not offered in most Rule based or weather model. Most existing tools require manual data entry or inspections because this function empowers the farmer to act as real-time data collector and the model learns and improve with image classification of improving model making community- driven adaptive system.

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List of Abbreviation

Abbreviation	Description
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
GCNNS	Deep convolutional neural networks
CNN	convolutional natural networks

1. Introduction

1.1 Background & literature review

Insect infestations remain the greatest challenge to paddy cultivation in Sri Lanka, causing severe losses in yield and financial hardships for the farmers annually. There life cycle is conduct with the climate changes, locations, diapauses like situations. So traditional approaches to forecasting such as seasonal calendars, field observations, and generalized warning remain incomplete, reactive, and non-responsive to the rapidly changing pest behaviors caused by climate variability. So this kind of traditional methods are not useful in the furthermore. This leads to delayed detection, overuse of pesticides, and low productivity. Current advances technology such as in Artificial Intelligence (AI), machine learning, and computer vision also now make it possible to build dynamic, data-driven pest forecasting systems that can analyze multiple environmental factors simultaneously Using real time data factors and reasons. AI models are also able to predict outbreaks more accurately earlier by integrating past data on pest outbreaks, weather, soil type, crop growth stages, and field images provided by farmers.

Having below kind of novelty features these component gave best of the distinguish it from the existing other than pest prediction tools:

1. **Localized of Sri Lanka-specific Modeling** – AI models designed on regional pest fauna, paddy crops, and monsoon-driven climate patterns, rather than generic global datasets.

2. **Multimodal Data Fusion** – Fusing meteorology data, soil parameters, historical pest data, and real-time farmer images to make accurate, location-specific predictions.

3. **Dynamic Learning** – The system continually updates predictions throughout the cultivation period based on real-time field data, accounting for unexpected climate fluctuations and pest behavior adjustments.

4. **Community-supported Data Collection** – Pest images and outbreak reports are submitted by farmers, gradually enhancing the accuracy of the system while fostering user participation.

5. **Stage-specific Actionable Recommendations** – Instead of simply predicting pest occurrence, the system provides targeted, crop-stage-based control measures, eliminating duplicative pesticide use and cost.

This synergism of local adaptation, constant learning, multimodal integration, and actionable guidance puts the system in a position to resolve Sri Lankan paddy farmer's actual-world pest management problems in a very unique manner from the technology world and having new technologies also.

1.2 Literature review

Pest attack forecasting has been a vital component of precision agriculture with tremendous potential in reducing yield loss, simplifying pesticide application, and enhancing environmental sustainability, particularly for weather-sensitive crops such as paddy. Traditional forecasting methods in Sri Lanka and the majority of developing nations rely mainly on fixed seasonal calendars, farmer experience, and manual scouting, which are inadequate in terms of providing early warnings or adapting to changing pest trends [1]. Meteorological conditions such as temperature, humidity, and rainfall have been discovered to greatly influence the dynamics of pest populations, triggering the application of time-series models such as Random Forests, as well as advanced deep learning methods such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for temporal pattern recognition [2].

"Wang at.el [3] introduced architecture such as let Net and Alex net to classify various crop pests. Besides, computer vision techniques specifically Convolutional Neural Networks have been trained to identify pests and pest-induced crop damage from field images using large datasets, even though there is still a problem of data scarcity, class imbalance, and the need for pest species localization [4]. "Image segmentation and other approaches have been used to preprocess the images, after which the features of target images are extracted. Then machine learning methods, such as random forest, k nearest neighbors, support vector machine (SVM) , etc are used to perform pest identification." [5] [6] [7] [8]. "Although the approaches described above have produced good identification results, they have flaws such as insufficient extracted information, manual identification of critical features, and low feature extraction efficiency. Deep convolutional neural networks (DCNNS) have become an effective method to solve the problems mentioned above in recent year" [9].

"Furthermore, deep learning technology based on convolutional natural networks (CNN) has made significant progress in identifying objects, such as pests. Due to the poor identification performance of CNNs in the field, [10] proposed data augmentation approach based on CNNs". [11] Image background removal was performed during data acquisition using the Gaussian mixture model to aid the classification and identification of insects in the images." [12]

The methods are based on saliency map and applied it to DCNNs for pest identification performance for CNNs and found that the accuracy was better than that previous one. These hybrid approaches, often based on attention mechanisms or ensemble learning, outperform single-source models and can issue dynamic, location-specific predictions rather than static seasonal alerts. However, most existing systems are customized to commercial agriculture in developed nations and are not suited to smallholder farming contexts in Sri Lanka, where pest diversity, rice variety-specific vulnerabilities, monsoon-driven climatic uncertainty, and limited rural internet connectivity significantly affect outcomes [13] Most models also fail at prediction by not converting outputs into actionable, stage- specific pest control recommendations. Key gaps in the literature include the absence of Sri Lanka-specific pest forecasting models tuned on local historical outbreak data, the paucity of lightweight offline-enabled AI apps for low-connectivity environments, and the absence of forecast integration with cost-sensitive intervention

suggestions. The traditional monitoring methods obtain pest information by regular field surveys. Although field survey can obtain detailed pest information, it is time consuming, labor-intensive, and limited in large scale surveys. [14] So having below advanced novelty features is very usefully and user friendly [15].

“Cheng et.al [16] used a deep residual network to identify to the 10 different categories of crop pests in complex background. In comparing with SVM and back propagation, their model performs better and achieved and accuracy rate of 98.67%.”

“Temperature-based risk maps are widely used to represent the ability to develop and establish in a given geographical area [17]. They are of particular interest to policymakers because they can be used in integrated pest management to predict how the biodemographic characteristics of a given species would change in response to temperature variation over large spatial scales (regional, national, or continental).Some tool have been developed to obtain such risk maps and ae generally made available to end users either as R packages [18] or as decision support systems software such as Maxent [19], [20] ILCYM [21] or CLIMEX [22] [23]”.

“The mathematical expression to produce maps that help to visualize the variability of risk at reginal and continental scales is usually linked to a climate database to extract the values of the environment variables ,defined as input for the risk assessment [24] [25]”.

Addressing these information and gaps using community-driven data curation, transfer learning for pest image classification, and climate-adaptive temporal modeling can enhance early detection accuracy and ease farmer adoption. In summary, the literature leads one to conclude that the best pest forecasting solutions for paddy cultivation must be localized, data-intensive, and farmer-focused, marrying predictive accuracy with actionable advisories that are driven by socio-economic ground realities.

1.3 Research GAP

Key Features	Research 1	Research 2	Research 3	Propose Research
Uses multimodal data (weather + soil+ seed type + crop stage + pest)	Weather only ☒	Weather only ☒	Weather only ☒	☑
Farmer Participation (Upload photos + Field data)	☑	☒	☒	☑
Adaptive / Continuous learning AI	☒	☒	☒	☑
Computer Vision for pest Identification	☒	☒	☒	☑
Real-time forecasting Mobile /web Alerts	☒	☒	☒	☑
Practical Farmer Decision support	☒	☒	☒	☑

1.4 Research problem

Pest attack forecasting is a critical component of modern agricultural systems for Sri Lankan paddy farmers, in which unexpected infestations of pests can devastate crops and livelihoods. The AI-Driven Integrated Agricultural Support Platform's Pest Attack Forecasting components aims to address this by applying state of the art machine learning techniques to predict pest infestations with high accuracy. By calculating past weather patterns, soil, and past infestation history, the system identifies patterns and trends reflecting possible infestations by pests.

As proactive as it can be, farmers can take countermeasures before infestations by pests happen and devastate crops in massive proportions, with minimum pesticide consumption and crop loss. The component is refreshed in real-time with reports of pest observations by farmers and hyper-local weather forecasts, soil type so its projections remain relevant throughout the season. Fixed rules or past data-based rule-based or seasonal forecasting systems, however, are based on pre-programmed rules or historical observations and fail to change according to changing environmental conditions or pest dynamics. For instance, it considers humidity levels, crop growth stages, and paddy seed varieties in order to forecast according to some context. Farmers are alerted by SMS or mobile notifications web notification in advance for having if near have pest attack, and they are able to apply proper interventions, Eg:- spraying with bio- pesticides or adjusting irrigation schedules like this. The incorporation of community-contributed data also enhances the precision of the system because farmers contribute images and field reports that enrich the AI models. Such co-operation not only increases prediction precision but also motivates farmers to maintain a knowledge-sharing culture.

By overcoming the disadvantages of conventional pest control protocols, the Pest Attack Forecasting module is a reflection of how AI can transform agriculture into a more robust, sustainable, and productive sector. Its novelty lies in how it can combine multimodal data inputs, adaptive learning, and farmer engagement to deliver precise, actionable insights that guard crops and optimize resource use. In total, this innovation advances the overall mission of the platform is empowering Sri Lankan paddy farmers with data-driven instruments that enhance productivity, profitability, availability and climate resilience according to the various of paddy type.

2. Objectives

2.1 Main Objectives

The primary objective of the Pest Attack Forecasting component is to establish an AI-driven predictive system for enhancing pest control for Sri Lankan paddy growers. According to past pest data, present environmental factors, and machine learning protocols, the system will issue early and accurate warnings of oncoming pest infestations. This early warning system tries to curtail crop loss, prevent excessive pesticide use, and maximize overall yield sustainability. In addition, the system introduces farmer-submitted data to make predictions more refined to the extent that the measures implemented are localized and also responsive to the stages of crop growth and territorial climatic patterns.

2.2 Specific Objectives

Timely correct prediction of Pest Outbreaks – Examining historical records of infestation alongside trend analysis of weather, crop type and soil in order to predict when infestations are to occur.

Real-time Alerts – Send SMS and mobile notifications web notification to farmers, hence equipping them with the capability to initiate prompt preventive measures on time.

Reduce Pesticide Reliance – Optimize pest management by recommending interventions only when necessary, lowering costs and environmental impact.

Adapt to Local field Conditions – Leverage farmer-submitted data Eg: pest photos, field observations to regularly refine prediction models for local variations.

Enhance Farmer Decision-Making – Make actionable recommendations on pest-resistant crops and optimal treatment timing to assist in optimizing harvest quality and profitability accurately.

Enhance and improve the usability issues: Refine the sentiment analysis models frequently in response to user comments and changing linguistic trends. Create for farmers feedback loops so the system may learn from user interactions and gradually improve its replies. In order to ensure a flawless and easy to use user experience, address technical difficulties and constraints.

Through such objectives, the Pest Attack Forecasting system aims to transform traditional reactive pest control into a clever, data-based process that is more efficient in agriculture

3. Methodology

The pest attack forecasting system employs a well-organized, data-oriented methodology to forecast pest attacks and enable farmers in Sri Lanka to make proactive decisions regarding their paddy crops. The methodology employs machine learning, real-time data analytics, and community feedback to assure accuracy, flexibility, and scalability.
The process begins with such as data acquisition, retrieving pest history records, climatic trends temperature, humidity, rain, soil health data, and crop growth stages from agronomic databases and weather API. Preprocessing is performed on the data to address missing values, feature scaling, and remove noise. Supervised learning models, Random Forest, Gradient Boosting, or Long Short-Term Memory (LSTM) networks, are trained to recognize correlations between environmental factors and pest infestations. The machine learning model automatically updates its predictions constantly using real-time field data from farmers, including pest reports and crop health images shared via a mobile app or website. Hyperlocal weather forecasts from weather stations add more accuracy. For usability, the system offers self-contained SMS and in-app alerts when it have the face the pest attack before it big to the full of field pest risk thresholds are exceeded, along with personalized countermeasures. It has a feedback loop for farmers to submit outcomes, facilitating continuous model tuning through reinforcement learning. This methodology ensures correct prediction despite changing climate trends and pest patterns, ultimately forming the basis for sustainable and efficient pest control for paddy culture.

Using Software Development life cycle we can describe it simply.

1. Planning:– What are the things we need to achieve in doing pest attack forecasting research and how to get data for the do our project and also how to this affective in real time.

2. Analysis:- We need to analysis for our finding data and what kind of software using be okay for our project.

3. Design:- Design the AI models ML algorithms using the time series , weather data , pest data and also paddy data . And also define what are the things should be input data and output information data and also start design UIUX field in system.

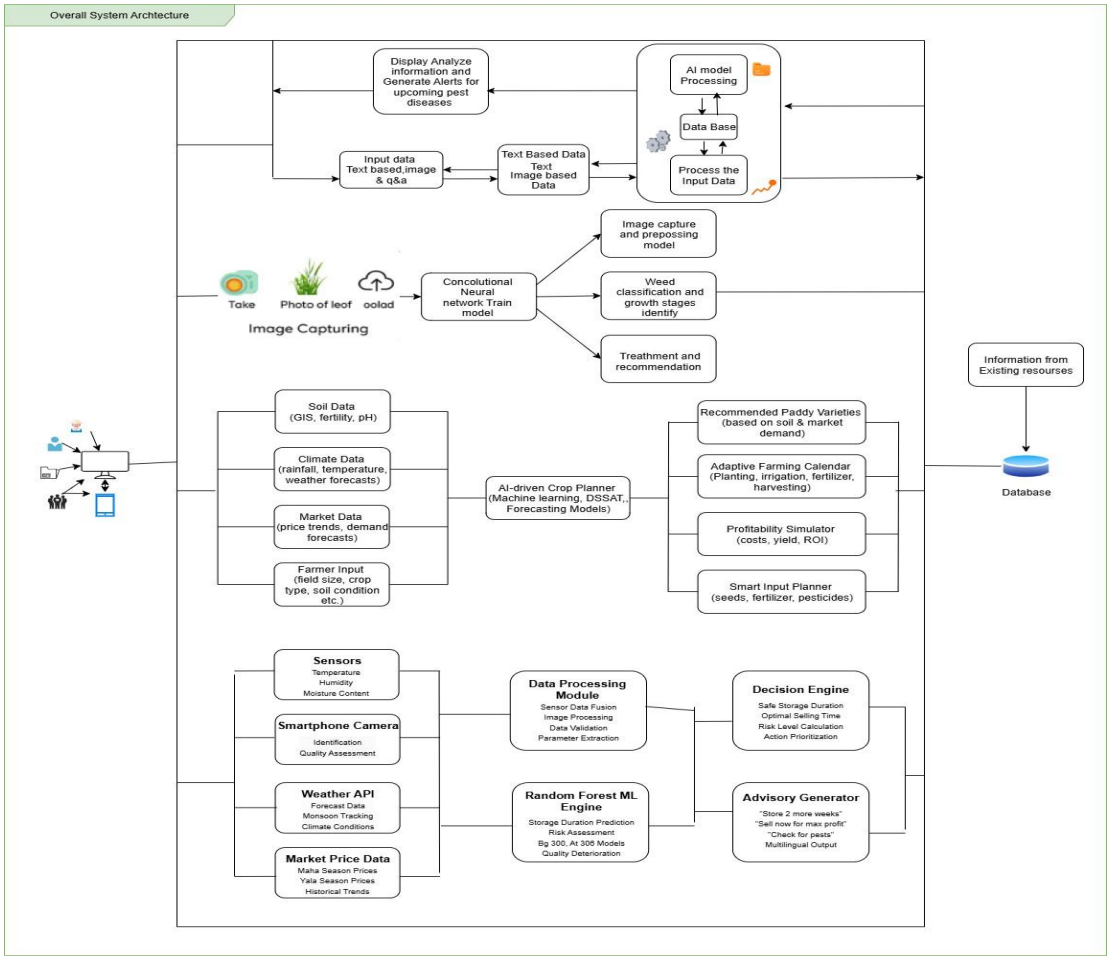
3.1.1 Software development life cycle

4. Implementation:- Collected data using and trained model using like LSTM for time series forecasting .Build backend for API connections for forecasting alerts . Also develop mobile web based system.

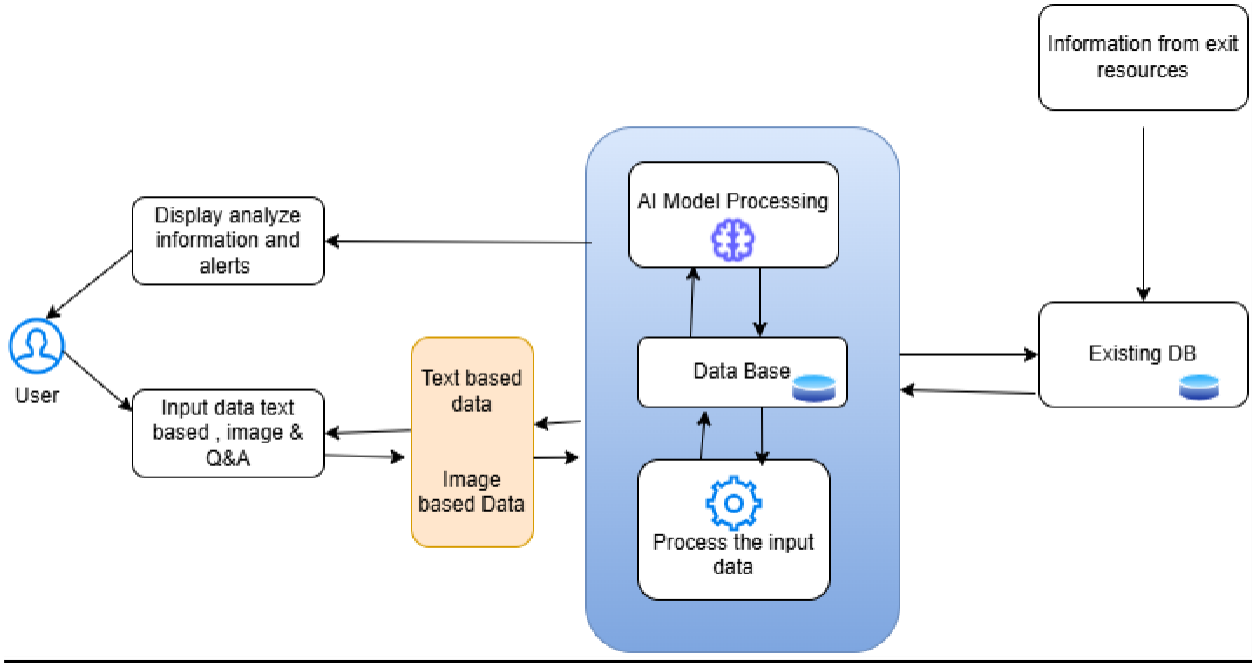
5. Testing and Integration:- In this phase we can ensure system works accurately before handover the client. We can used for this testing type like automation testing, performance testing, API testing, unit testing methods. After all testing we can have finally bugs free validate system.

6. Maintains:- Keep system update with real time and improve a ML model using updated data.

3.1 System Architecture



3.1.2 Overall System Architecture



3.1.3 Pest Attack Forecasting system architecture

3.2 Data Collection

Development depends on multimodal data such as weather patterns, soil health records, pest/weed incidence history, crop images, farmer feedback, and satellite data. Data on local market prices, transport costs, and farmer demographics are also vital for contextual relevance. Continuous collection and integration of farmer-uploaded data e.g., pest images, harvest logs will fuel active learning models, while GIS-based land profiling and real-time weather API input support precision agriculture. All data collection must comply with ethical standards, including farmer privacy and informed consent.

3.3 Software Solutions

- AI model TensFlow, LSTM, CNN, NLP
- GEO pandas for risk map
- Weather API
- Web Scrapping
- NOSQL database
- React native, React JS
- Taliwind CSS, Boostrap
- Chart.js, D3 js
- Node.js
- Epress.js

- Cloud Services
- GIT hub/ GIThub Flow
- Selenium
- Postman

3.4 Commercialization

The proposed pest attack forecasting having many of high commercialization due to the growing challenges of the farmers faced of managing and controlling pest attack. At present farmers are only used manual observation and delayed of the agriculture report often lead to the significant crop damages.

Target Market: Primary Farmers and new farmers in Sri Lanka who cultivate paddy and system can extended to the agriculture officers, cooperatives and having agri Business Company.

Value proposition: The system allows to the take prevention action before take pest attack, thereby reducing crop losing, lowering pesticide costs and improve harvest quality. Unlike having existing traditional methods, this solution provides the real-time data driven prediction.

3.5 Future Scope

Scaling Up to several of Other Crops can scale up pest prediction models to other crops in Sri Lanka such as maize, tea, coconut, and vegetables to make the platform more useful for a greater community of farmers.

AI-Powered Pest Control Automation can link forecasting results to robotic spraying systems or precision application drones, allowing instant, directed pesticide application in high-risk areas.

Voice-Enabled Assistance can implement AI-based voice assistants in Sinhala and Tamil to allow farmers to use the system without using hands, enhancing accessibility for older or low-literacy users.

Block chain-Based Pest Outbreak Report Verification can apply block chain technology for secure verification and storage of pest outbreak reports to offer transparency to agricultural authorities and stakeholders.

Interoperability with Government Agri-Programs can interoperate the system with Sri Lanka's Department of Agriculture databases and subsidies so that farmers are able to access prescribed pesticides or biological controls at subsidized prices real-time.

Economic Impact Estimation can enhance the system not only to predict pest threats but also predict possible yield loss and economic damage to allow farmers to make well-informed investment decisions.

4. Project Requirements

4.1 Functional Requirements

- Pest Image upload
- Weather & soil Data Collection
- Alerts Generation
- Pest Risk Prediction
- View Pest History
- Actionable Recommendations for Pest stages and also Paddy type
- AI Model updating
- Role-Based Access
- Storage

4.2 Non-Functional Requirements

- Performance
- Availability
- Scalability
- Usability
- Reliability
- Maintainability
- Capability
- Localization
- User Experience

4.3 User Requirements

- Input Pest basic crop detail
- Input weather details
- Receive timely Pest attack alerts & recommended actions
- Upload Pest Image
- View history of the past attacks and outbreaks for get action
- Use system in preferred language
- Trust data privacy

4.4 System Requirements

- Mobile or Desktop
- Backend With AI/ML model deployment
- Weather API
- Database

4.5 Use Case Diagram

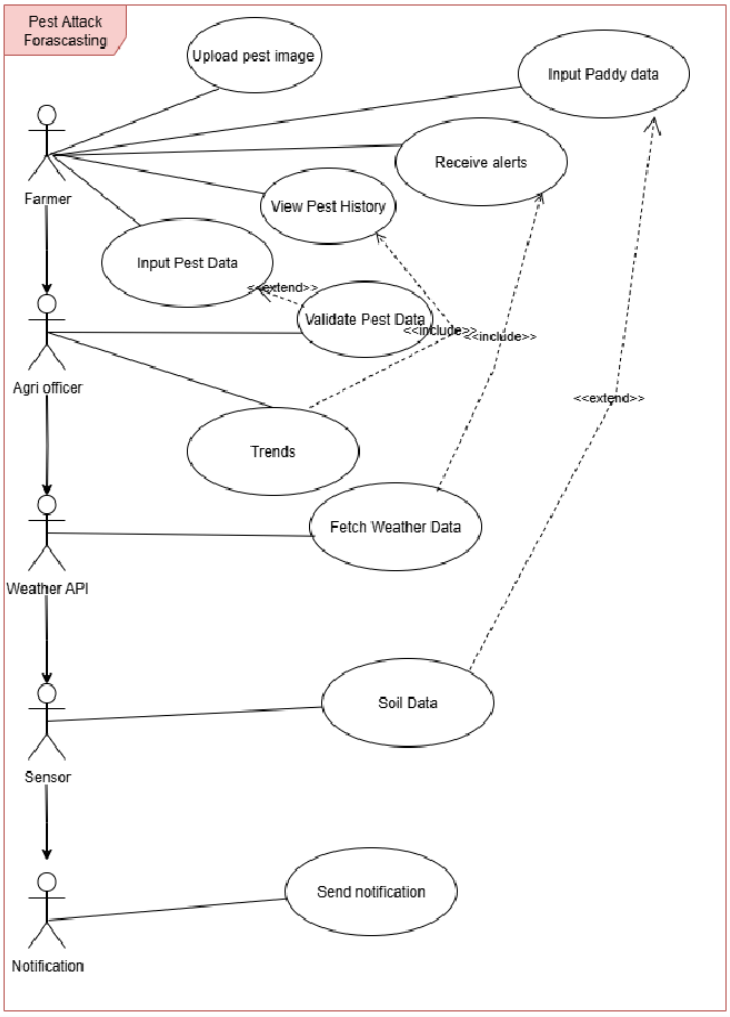


Figure 4 Use case diagram

4.6 Test Cases

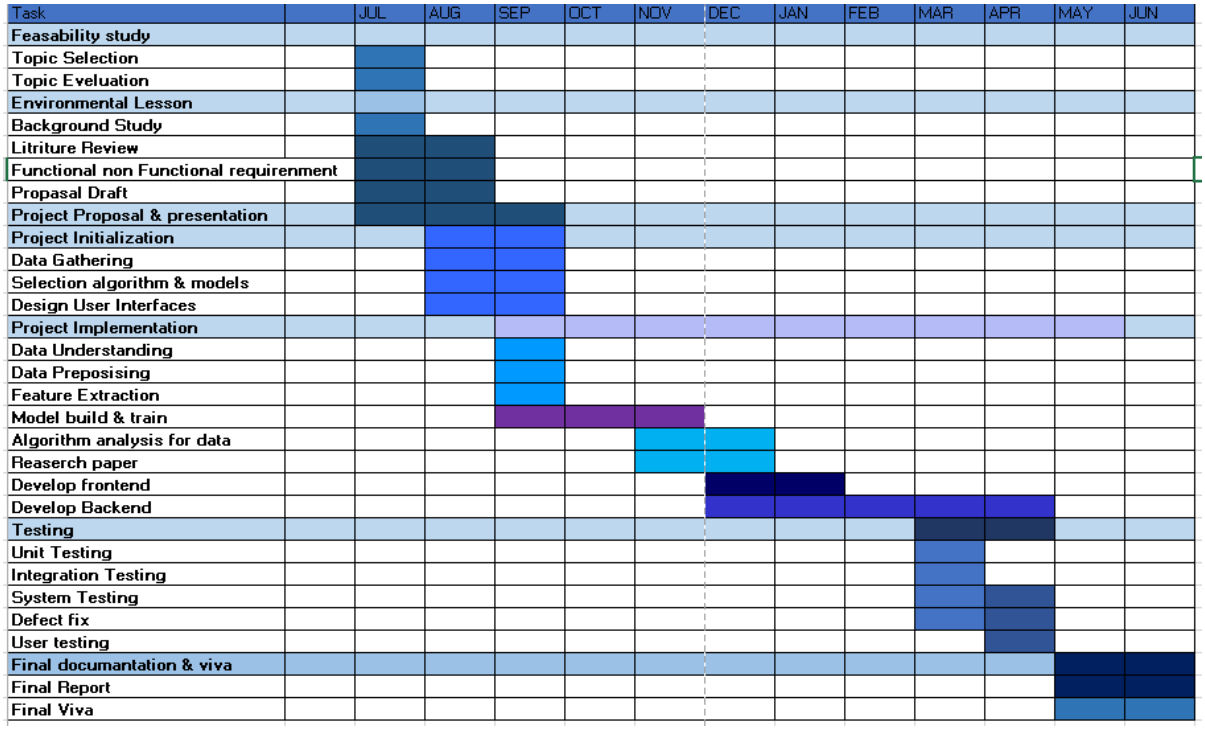
Test Case ID	Description	Preconditions	Test Steps	Expected Result
01	Verify the pest Image Upload	User logging to the system	1. Navigate to the pest forecasting page. 2. Click the "Upload Image " 3. Upload Image 4. Submit form	1. Receive Image successfully upload massage. 2. Start processing 3. Identify pest and show details.
02	View the notifications	Logging to the system	1. Navigate the notification center 2. Click the pest forecasting button 3. Show forecast notification	1. Open notification 2. Show details 3. Can Delete notification

03	View forecasting history	Logging to the system	1. Navigate to the Pest forecasting page 2. Click the "Pest History " Button 3. Select The Date Range	1. Show the pest history details including the Pest type solution and paddy type
04	Send the Pest Alerts notification	1. Identify the threads	1. See the weather map 2. Model processing with pest history 3. Also paddy type and paddy age	1. Send the notification for user including pest attack type and recommendations

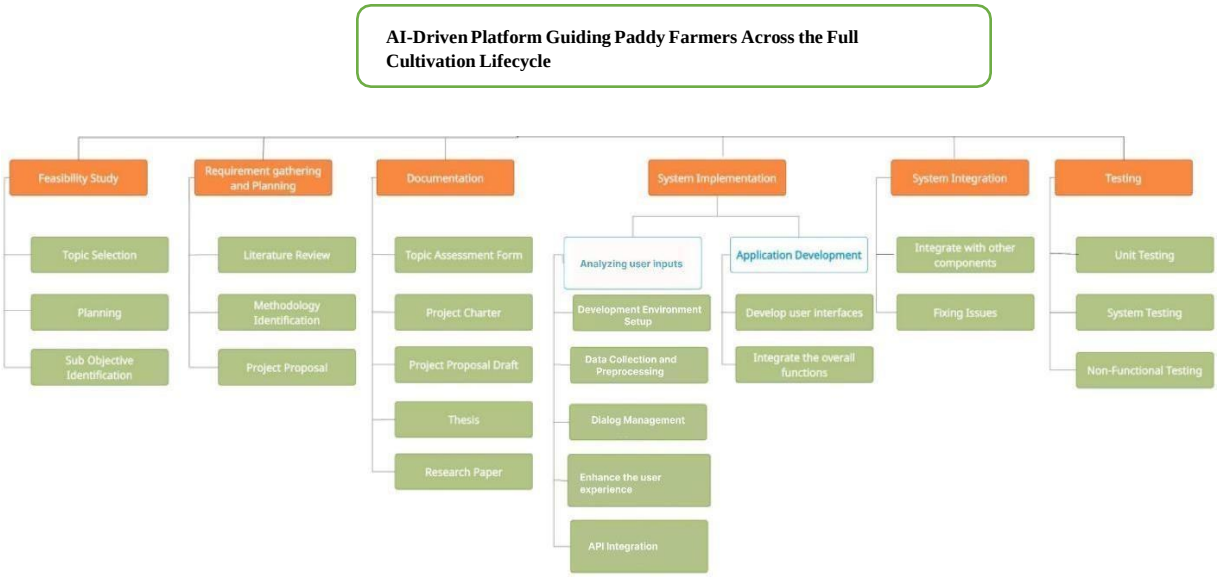
5. Budget Planning Justification

Description	Cost
Electricity	3000.00
Internet	10000.00
Communication	4000.00
Transportation	5500.00
Paper Publishing	25000.00
Backend Hosting	25000.00
Software Purchasing	7000.00
Total	70500.00

6. Gantt Chart



7. Work Breakdown



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9. Appendix

9.1 Wireframes

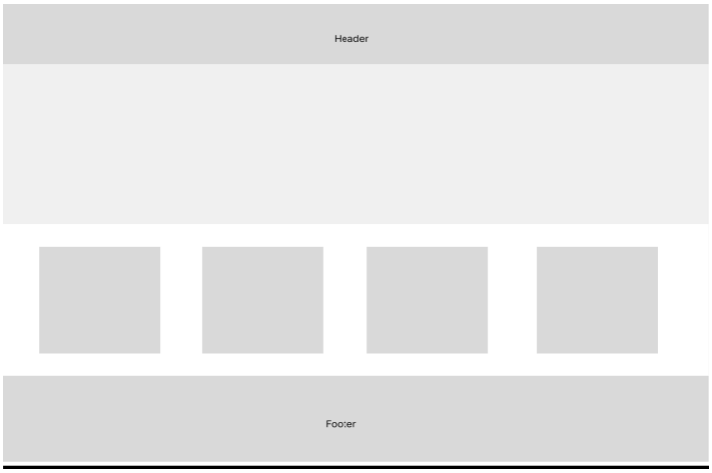

Figure 2.1 Homepage of pest attack forecasting

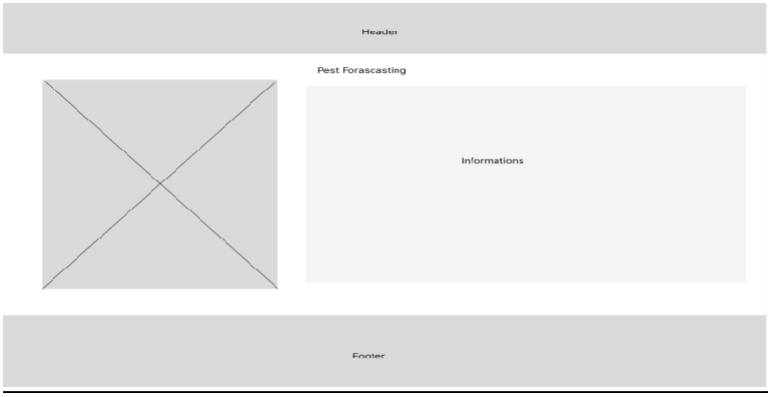

Figure 2. 2 Information of pest

Figure 2.3 Input Crop details

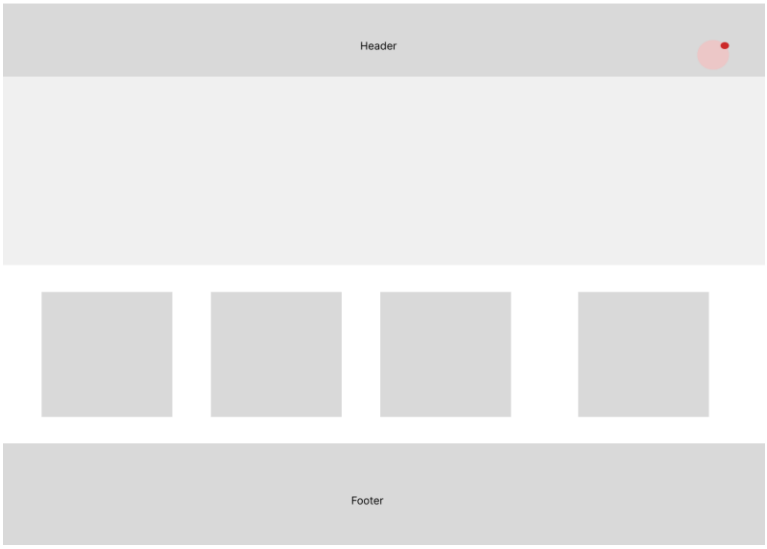

Figure2.4 Notifications receive Showing

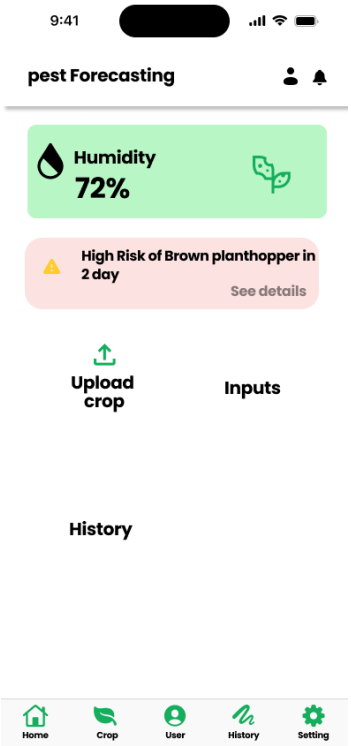

Figure 2.5: app version

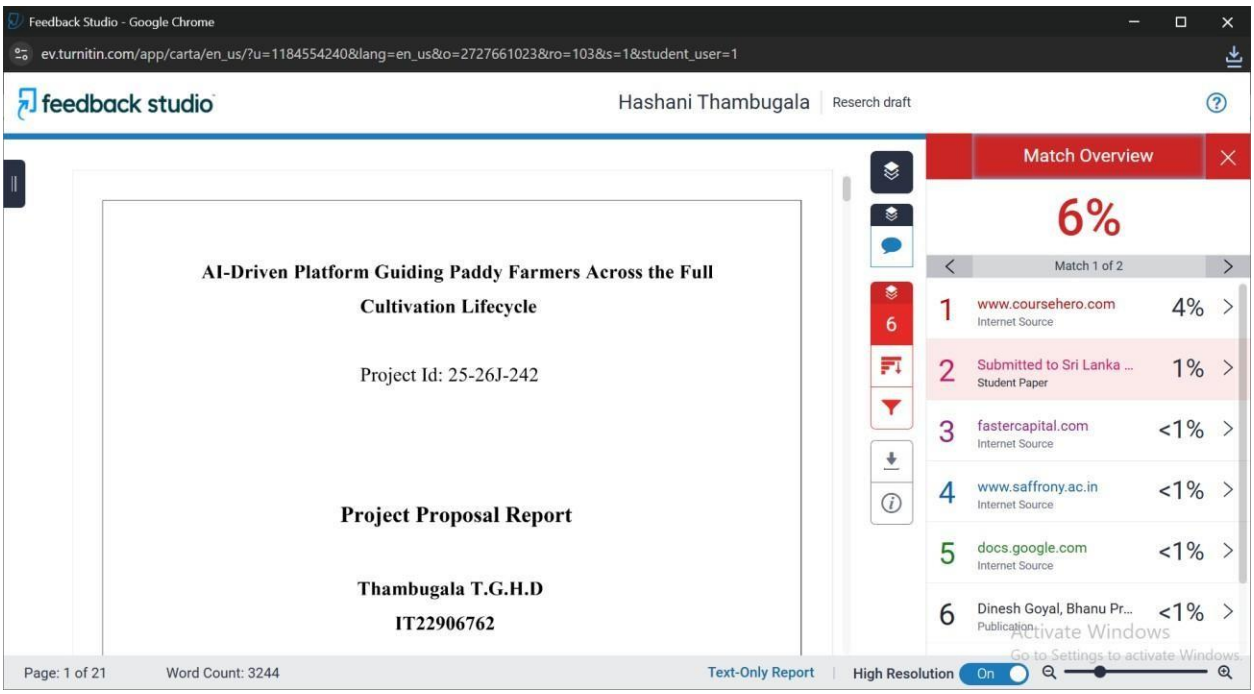

Figure 5.d plagiarism checker result